

NATIONAL OPEN DATA INNOVATION CHALLENGE 2 — ENERGY & ENVIRONMENT

PROBLEM STATEMENT DEFINITION — GridIQ

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Data honesty legend

- **REAL** — confirmed open dataset, verified in the OIC2 hub (available now)
- **DEMANDED** — does not exist as open data; we formally call for its release
- **ASSUMED** — benchmark/model assumption where no Botswana data exists (stated as such)
- **EXTERNAL** — published by BPC/government but not held in the hub (cited directly)

Health and environmental burden is stated as **motivation, not as a finding**: no air-quality monitoring data for Morupule exists in any open source, and we do not quantify it.

Main problem statement

Botswana's electricity system is planned as a series of separate projects rather than as one interconnected system. The result is over-reliance on coal-fired generation at Morupule, continued dependence on imported power, a rural population still waiting for the last mile, and a network operated without visibility of where it is constrained or which assets are critical.

The people who feel this are Botswana households and businesses (unreliable, expensive power) and the BPC and Ministry planners who must choose what to build next (no system-cost view to choose with). The result is expensive, unreliable and inequitable power, and investment that optimises individual resource cases instead of total system cost.

Problem statement 1 — Supply is planned project-by-project, so it cannot reliably or cheaply cover demand

Generation still cannot cover demand at home, and the new supply that would fix that is chosen without seeing the grid. Two coal stations — Morupule A (132 MW) and Morupule B (600 MW) — plus diesel peaking dominate the fleet, with installed capacity around 892 MW against peak demand of 678 MW (2024). On a five-year rolling average domestic generation has covered only about 62% of supply; the rest is imported, and in Q1 2026 80.9% of imports came from Eskom alone, at high and volatile cost. Local generation has covered as little as ~44% of distribution in a quarter.

At the same time, each new project — a solar farm, a coal plant, a battery site — is appraised on its own resource case (good sun, cheap coal reserves), not on what it costs the whole system once transmission capacity, congestion and storage are counted. Solar resources sit far from load, so a strong site can force a costly line upgrade that the resource map alone never shows.

Who is affected: households and industry (unreliable, expensive power); BPC and Ministry planners (no system-cost view to choose with); the budget and tariff payers (grid spend that a better-placed project would not have needed).

Cause: the first least-cost integrated resource plan is not due until 2026, so generation, storage and grid are decided as separate projects — with one dominant generation site and one dominant import corridor. This is also why “more coal AND more renewables AND fewer imports” only works as one plan, not three.

Datasets used to validate this problem

Dataset	Source	Type	How it validates
Electricity generation & distribution by source, quarterly back to 2016	Statistics Botswana	REAL	Q1 2026 = 90.8% coal, 9.1% solar; local generation covered 80.4% of distribution (best quarter on record); series shows quarters as low as ~44%.
Africa Energy Balance 2025	AFREC, African Union	REAL	2023 national balance: whole-economy import dependence, plant efficiency ~25%, delivery losses ~13.8%.
BPC Integrated Annual Report 2023	Botswana Power Corporation	EXTERNAL	Publishes T&D losses of 642 GWh in FY2023 (14.51% of units); historical series 509–676 GWh. Our loss anchor.
OpenInfraMap & Global Power Plant Database	OpenInfraMap / WRI	REAL	Existing Botswana plants with capacity and fuel type, plus transmission-grid geometry — the fleet and grid we are optimising.
Southern African Power Pool (SAPP)	SAPP	REAL	Published load data and interconnection-capacity reports across the regional grid — the alternative corridors.
Eskom Data Portal (South Africa)	Eskom	EXTERNAL / regional benchmark	Quantifies the single-corridor import exposure.
Global Solar Atlas, Global Wind Atlas & NASA POWER	World Bank / DTU; NASA	REAL	Resource rasters and point series for candidate siting — the resource case that must be tested against grid cost.
IRENA & Regional Renewable-Investment Data	IRENA	REAL	Capacity and cost benchmarks for the optimisation.
Grid infrastructure & demand data	Botswana Power Corporation	DEMANDED	No BPC dataset confirmed available. We call for line ratings, impedances and substation loads to be released.
Grid generation statistics & average demand figures	BOCRA	DEMANDED	No public dataset available.

Dataset	Source	Type	How it validates
Business register data on energy-sector / large industrial consumers	Statistics Botswana	DEMANDED	No public dataset identified — needed to size industrial demand.
Line ratings/impedances; capex per MW and per km	Engineering references / regional benchmarks	ASSUMED	Where no Botswana cost data exists, use published benchmarks and label them.

Proposed conceptual solution

System-wide optimisation engine (Engine 1) and flow, congestion & loss layer (Engine 2). One digital model of generation, storage, grid capacity and demand that jointly answers what to build, where, how much and how to connect — minimising imports and total system cost, instead of judging each project on its own resource case. A linearised DC power-flow / OPF model over the real topology shows where flow concentrates, which constraints bind, loss per line, and the redispatch cost when a constraint forces costly supply. (Note: power is not “routed” — current divides across all parallel paths by impedance; we model constraints, not shortest paths.)

The engine’s loss output is reconciled against BPC’s own published figure: **642 GWh lost in FY2023 (14.51% of units), BPC Integrated Annual Report 2023** — independently cross-checked by the World Bank (T&D losses 15.35% as of 2021) and AFREC (~630 GWh, 13.8% of supply, 2023). Scope note carried openly: BPC’s figure covers transmission *and* distribution, and Statistics Botswana’s series excludes network losses entirely, so the model reports its transmission-level result as a bracket against BPC’s T&D total.

Problem statement 2 — The network is operated reactively

Without a published view of where the network is congested, or which assets have no redundant path, faults are answered after they happen rather than anticipated. No asset age, condition, inspection or failure history is published for BPC or any Botswana utility.

Who is affected: households and businesses that experience outages; BPC crews who respond instead of pre-empt.

Cause: no criticality or condition data. A prediction model is impossible without a training signal that does not exist.

Datasets used to validate this problem

Dataset	Source	Type	How it validates
OpenInfraMap (OSM transmission grid)	OpenInfraMap / OSM	REAL	Topology alone supports criticality: which assets have no redundant path and would cause an outage, not just a cost.
Electricity generation & distribution by source, quarterly	Statistics Botswana	REAL	Observed generation/throughput swings that indicate system stress.

Dataset	Source	Type	How it validates
Southern African Power Pool (SAPP)	SAPP	REAL	Interconnection-capacity reports — where regional backup could relieve local stress.
Grid infrastructure & demand data	Botswana Power Corporation	DEMANDED	Asset condition, ratings, inspection and failure history absent.
Asset age / condition / failure history	—	DEMANDED	No public source exists for any Botswana utility.

Proposed conceptual solution

Asset-risk screening layer, with a data contract (Engine 3). A ranked watch-list of lines, substations and transformers by structural criticality and estimated stress, computed from topology alone. Shipped with a one-page data contract listing the exact BPC fields (age, condition, inspections, failure history, ratings, impedances) that would upgrade the proxy score into a real predictive model. Honest ask, not an overclaim.

Problem statement 3 — The last mile is stuck

As of October 2024, 463 of Botswana’s 565 gazetted villages were connected, leaving 102 unconnected — roughly one in six. Grid extension to these remaining villages is disproportionately expensive; they are exactly the low-density, far-flung settlements skipped in earlier phases. National access sits at 73.9% (NEUS 2022/23) / 76% (Africa Energy Portal), falling to 14.3% in Ngwaketse West. Government’s stated fix is solar off-grid electrification, treated as the default answer everywhere rather than chosen per village.

Who is affected: rural households, and students like Gomolemo — issued a government laptop under the School Digitisation programme, in a village with no power to charge it. A laptop reached him before the grid did. (*Representative scenario based on documented evidence; not a documented individual.*)

Cause: no systematic per-village comparison of solar, wind, biomass or hybrid against the cost of extending the line.

Datasets used to validate this problem

Dataset	Source	Type	How it validates
National Energy Use Survey Main Report 2022/23	Statistics Botswana	REAL	73.9% national access, 14.3% Ngwaketse West; total national consumption 43,001.7 TJ/yr (66% biomass), by region and fuel.
Africa Energy Portal: Botswana Country Page	AfDB / AUDA	REAL	76% national electricity access; 15%/36%/50% renewable targets for 2030/2036/2040.

Dataset	Source	Type	How it validates
2022 Population & Housing Census	Statistics Botswana	REAL	2,359,609 persons, district-level breakdowns — demand and access denominators.
Administrative & Census Sub-District Boundaries (shapefile)	Statistics Botswana GIS	REAL	32 validated sub-district polygons for spatial joins.
Meta High-Resolution Population Density	Meta / HDX	REAL	30 m population density — settle demand where people actually are.
OpenStreetMap / HDX Botswana Extracts	OpenStreetMap / HDX	REAL	~1.5 M mapped buildings, roads, water sources and markets — the settlement layer for per-village scoring.
Global Solar Atlas & NASA POWER	World Bank / DTU; NASA	REAL	Per-site irradiance and time series for solar sizing; evening-load mismatch drives solar-plus-storage.
Agricultural Census 2025 (livestock) / FAO livestock production	Statistics Botswana / FAO	REAL	1.64 M cattle nationally — district biogas feedstock potential.
Tracking SDG7: Energy Progress Report	World Bank / ESMAP / IEA / IRENA / WHO	REAL	Access and clean-cooking indicators for national context.
BPC rural electrification / SE4ALL gap analysis	BPC / SEforALL	REAL + DEMANDED	Gazetted vs connected status; live connection records would sharpen targeting.

Proposed conceptual solution

Community-level fit layer — VillageFit (Engine 4). For each still-unconnected village, score solar, wind, biomass, biogas or hybrid against that village’s own resource data, distance from the grid, and load profile — including how the village actually uses power (e.g. evening study). Where a line is not the cheapest way to serve a settlement, recommend a right-sized mini-grid instead. Every recommendation traces back to a named dataset.

How the platform answers all three

GridIQ is **one shared network model** with four engines reading the same topology, demand and injection points:

Engine	What it addresses	Problem
Engine 1 — System-wide optimisation	Joint siting, sizing and connection against one objective	Problem 1
Engine 2 — Flow, congestion & losses	Binding constraints, loss per line, redispatch cost (the falsifiable wedge)	Problem 1
Engine 3 — Asset-risk screening + data contract	Structural criticality and proxy risk	Problem 2
Engine 4 — VillageFit community-level fit	Per-village technology fit vs line extension	Problem 3

Because all four read the same substrate, no recommendation can silently contradict another — this is what makes it a platform, not four scripts.